^{177}Lu-PSMA-617 versus a change of androgen receptor pathway inhibitor therapy for taxane-naive patients with progressive metastatic castration-resistant prostate cancer (PSMAfore): a phase 3, randomised, controlled trial

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Summary

Background [^{177}Lu]Lu-PSMA-617 (^{177}Lu-PSMA-617) prolongs radiographic progression-free survival and overall survival in patients with metastatic castration-resistant prostate cancer previously treated with androgen receptor pathway inhibitor (ARPI) and taxane therapy. We aimed to investigate the efficacy of ^{177}Lu-PSMA-617 in patients with taxane-naive metastatic castration-resistant prostate cancer.

Methods In this phase 3, randomised, controlled trial conducted at 74 sites across Europe and North America, taxane-naive patients with prostate-specific membrane antigen (PSMA)-positive metastatic castration-resistant prostate cancer who had progressed once on a previous ARPI were randomly allocated (1:1) to open-label, intravenous ^{177}Lu-PSMA-617 at a dosage of 7·4 GBq (200 mCi) ± 10% once every 6 weeks for six cycles, or a change of ARPI (to abiraterone or enzalutamide, administered orally on a continuous basis per product labelling). Crossover from ARPI change to ^{177}Lu-PSMA-617 was allowed after centrally confirmed radiographic progression. The primary endpoint was radiographic progression-free survival, defined as the time from randomisation until radiographic progression or death, assessed in the intention-to-treat population. Safety was a secondary endpoint. This study is registered with ClinicalTrials.gov (NCT04689828) and is ongoing. In this primary report of the study, we present primary (first data cutoff) and updated (third data cutoff) analyses of radiographic progression-free survival; all other data are based on the third data cutoff.

Findings Overall, of the 585 patients screened, 468 met all eligibility criteria and were randomly allocated between June 15, 2021 and Oct 7, 2022 to receive ^{177}Lu-PSMA-617 (234 [50%] patients) or ARPI change (234 [50%]). Baseline characteristics were mostly similar between groups; median number of ^{177}Lu-PSMA-617 cycles was 6·0 (IQR 4·0–6·0). Of patients assigned to ARPI change, 134 (57%) crossed over to receive ^{177}Lu-PSMA-617. In the primary analysis (median time from randomisation to first data cutoff 7·26 months [IQR 3·38–10·55]), the median radiographic progression-free survival was 9·30 months (95% CI 6·77–not estimable) in the ^{177}Lu-PSMA-617 group versus 5·55 months (4·04–5·95) in the ARPI change group (hazard ratio [HR] 0·41 [95% CI 0·29–0·56]; p<0·0001). In the updated analysis at time of the third data cutoff (median time from randomisation to third data cutoff 24·11 months [IQR 20·24–27·40]), median radiographic progression-free survival was 11·60 months (95% CI 9·30–14·19) in the ^{177}Lu-PSMA-617 group versus 5·59 months (4·21–5·95) in the ARPI change group (HR 0·49 [95% CI 0·39–0·61]). The incidence of grade 3–5 adverse events was lower in the ^{177}Lu-PSMA-617 group (at least one event in 81 [36%] of 227 patients; four [2%] grade 5 [none treatment related]) than the ARPI change group (112 [48%] of 232; five [2%] grade 5 [one treatment related]).

Interpretation ^{177}Lu-PSMA-617 prolonged radiographic progression-free survival relative to ARPI change, with a favourable safety profile. For patients with PSMA-positive metastatic castration-resistant prostate cancer who are being considered for a change of ARPI after progression on a previous ARPI, ^{177}Lu-PSMA-617 may be an effective treatment alternative.

Funding Novartis.

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Published Online
September 15, 2024
https://doi.org/10.1016/S0140-6736(24)01653-2

See Online/Comment
https://doi.org/10.1016/S0140-6736(24)01919-6

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Introduction

Mortality remains high among patients with metastatic castration-resistant prostate cancer, despite the availability of multiple life-prolonging therapy options.[1] Androgen receptor pathway inhibitors (ARPIs) have been the most common first-line systemic therapy for the disease.

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See Online for appendix 1

Research in context

Evidence before this study

[^{177}Lu]Lu-PSMA-617 (^{177}Lu-PSMA-617) is a prostate-specific membrane antigen (PSMA)-targeted radioligand therapy for patients with PSMA-positive metastatic castration-resistant prostate cancer who have previously received androgen receptor pathway inhibitor (ARPI) and taxane therapy. We used the term "(metastatic castration-resistant prostate cancer) AND (lutetium) AND randomized NOT review[pt]" to search PubMed for English language publications from Jan 1, 2014 to May 13, 2024. We identified three completed and published randomised controlled phase 2 clinical trials and the pivotal phase 3 VISION trial. One of the phase 2 trials (NCT03392428) reported higher prostate-specific antigen (PSA) response and fewer grade 3 or 4 adverse events with ^{177}Lu-PSMA-617 compared with cabazitaxel in 200 patients who had previously received docetaxel. Another of the phase 2 trials (NCT04419402) reported that PSA progression-free survival was longer with ^{177}Lu-PSMA-617 plus enzalutamide versus enzalutamide alone as first-line therapy in 162 patients with metastatic castration-resistant prostate cancer. A third phase 2 trial (CTRI/2019/12/022282) compared ^{177}Lu-PSMA-617 with docetaxel pre-chemotherapy but the sample size was very small. The VISION trial (NCT03511664) showed that addition of ^{177}Lu-PSMA-617 to standard of care improved radiographic progression-free survival and overall survival compared with standard of care alone in patients with progressive, extensively pre-treated metastatic castration-resistant prostate cancer.

Added value of this study

To our knowledge, PSMAfore is the second phase 3 clinical trial of ^{177}Lu-PSMA-617 in patients with prostate cancer, and is the first phase 3 trial to assess the efficacy and safety of ^{177}Lu-PSMA-617 in the taxane-naive metastatic castration-resistant prostate cancer population. The results of the primary analysis presented here show that ^{177}Lu-PSMA-617 is an effective alternative to a change of ARPI for delaying disease progression and deterioration of quality of life in patients with metastatic castration-resistant prostate cancer who have initially progressed on an ARPI and for whom it was considered appropriate to delay taxane-based chemotherapy. The results of PSMAfore complement those of VISION, which assessed ^{177}Lu-PSMA-617 in the post-taxane metastatic castration-resistant prostate cancer population.

Implications of all the available evidence

The findings of PSMAfore expand the high-quality body of evidence for the efficacy and safety of ^{177}Lu-PSMA-617 in patients with metastatic castration-resistant prostate cancer in extending radiographic progression free survival. The previous phase 3 VISION trial and randomised controlled phase 2 trials mentioned above provided evidence for the efficacy of ^{177}Lu-PSMA-617 in patients previously treated with both ARPIs and taxane-based chemotherapy. PSMAfore now provides evidence that ^{177}Lu-PSMA-617 is a more effective therapy option with a better safety profile than a change of ARPI for disease control, as shown by prolonged radiographic progression-free survival, in patients for whom it is considered appropriate to change ARPI and delay taxane-based chemotherapy.

However, as ARPIs are increasingly initiated in earlier disease settings, the first-line therapy landscape for the condition is evolving.[2–6] For patients who have disease progression on a first-line ARPI, selection of the next appropriate therapy can be controversial. Evidence has shown that taxane-based chemotherapy is the most active therapy in this population;[7,8] thus, taxane therapy after initial progression on an ARPI is recommended by many clinical guidelines.[9–11] However, real-world data indicate that many patients and physicians select alternative therapies.[2–5,12] Reasons for deferring or avoiding taxanes include a desire by the patient or the physician to avoid side-effects that can negatively impact quality of life,[13,14] an assessment that the therapeutic benefit does not outweigh its negative side-effects, or a patient health status that renders taxanes medically inappropriate. As a result, many patients receive alternative therapies.[15–17] A change of ARPI is a common second-line therapeutic strategy among patients for whom it is considered appropriate to delay taxane-based chemotherapy.[2–5,7,18–21] Alternatives include poly(ADP-ribose) polymerase (PARP) inhibitors for patients with homologous recombination repair (HRR) gene alterations,[22–24] sipuleucel-T for asymptomatic or minimally symptomatic disease,[25] pembrolizumab for patients with microsatellite instability-high or mismatch repair-deficient tumours,[26] and ^{223}Ra for metastatic bone disease.[25] However, a minority of patients receive such therapies, and there remains an unmet need for therapeutic options for patients who wish to defer taxane-based chemotherapy.

[^{177}Lu]Lu-PSMA-617 (^{177}Lu-PSMA-617; also known as lutetium vipivotide tetraxetan) is a prostate-specific membrane antigen (PSMA)-targeted radioligand therapy approved for PSMA-positive, progressive, metastatic, castration-resistant prostate cancer following both ARPI and taxane therapy.[27–30] PSMA is often overexpressed on prostate cancer cells but has restricted expression in most normal tissues, enabling ^{177}Lu-PSMA-617 to deliver β-particle radiation specifically to prostate cancer cells and the surrounding tumour microenvironment, while minimising off-target effects.[31,32] The VISION trial showed the efficacy of ^{177}Lu-PSMA-617 in patients with metastatic castration-resistant prostate cancer who had received taxanes, with significant prolongation of imaging-based

(ie, radiographic) progression-free survival and overall survival following addition of ^{177}Lu-PSMA-617 to the standard of care versus standard of care alone.[33,34] In the phase 2 TheraP trial, ^{177}Lu-PSMA-617 was associated with improved intermediate endpoints, including prostate-specific antigen response, when compared with cabazitaxel for patients with metastatic castration-resistant prostate cancer who had received docetaxel, with no difference in overall survival.[35,36]

PSMAfore is a phase 3 trial that was designed to compare ^{177}Lu-PSMA-617 with a change of ARPI in taxane-naive patients with metastatic castration-resistant prostate cancer who had progressed once on ARPI as their most recent therapy, and were considered candidates for a change of ARPI. In this Article, we report the primary endpoint of radiographic progression-free survival, the planned third interim analysis of the key secondary endpoint of overall survival, and several other secondary and exploratory endpoints.

Methods

Study design

PSMAfore is an open-label, phase 3, randomised, controlled trial conducted at 74 sites (25 in North America and 49 in Europe; appendix 1 pp 2–5), investigating the efficacy of ^{177}Lu-PSMA-617 compared with a change of ARPI (to abiraterone or enzalutamide) in taxane-naive patients with PSMA-positive metastatic castration-resistant prostate cancer who had progressed once on a previous ARPI.

The study was conducted in accordance with the Declaration of Helsinki principles and Good Clinical Practice guidelines. Independent ethical review boards at all sites approved the protocol (appendix 2). All patients provided written informed consent.

The trial was sponsored by Novartis and designed and overseen by a steering committee comprising the lead trial investigators and sponsor representatives. During the trial, the steering committee monitored safety data and compliance data, including protocol deviations. Data were analysed by the sponsor and interpreted in collaboration with the lead investigators. All authors had full access to the data, vouch for the fidelity of the trial to the protocol, and critically reviewed and approved the manuscript. An independent data monitoring committee assessed the progress of the trial, safety data, and critical efficacy variables.

This trial is registered with ClinicalTrials.gov (NCT04689828) and is ongoing.

Participants

Eligible patients were aged 18 years or older, had PSMA-positive, metastatic, castration-resistant prostate cancer, and were candidates for a change of ARPI following one instance of disease progression on abiraterone, enzalutamide, darolutamide, or apalutamide as their most recent anticancer therapy. Candidacy for change of ARPI was determined by the treating physician during the informed consent discussion and was based on the patient's clinical history. Patients were ineligible if they had received chemotherapy for castration-resistant or hormone-sensitive prostate cancer (six or fewer cycles of adjuvant or neoadjuvant taxane therapy ≥12 months ago was allowed), immunotherapy (except sipuleucel-T) or systemic or hemibody radiotherapy (≤6 months ago), or were known to have actionable genomic alterations (eg, *AR-V7* or *BRCA1/2* alterations) for which alternative standard therapies were available (eg, PARP inhibitors). An Eastern Co-operative Oncology Group (ECOG) performance status of 0 or 1 and adequate organ function were required.

The criteria for PSMA-positive, metastatic, castration-resistant prostate cancer on centrally read baseline [^{68}Ga]Ga-PSMA-11 (^{68}Ga-PSMA-11) PET-CT scans required patients to have at least one PSMA-positive metastatic lesion (^{68}Ga-PSMA-11 uptake greater than in liver parenchyma), and no PSMA-negative lesions exceeding specified minimum dimensions. All lymph nodes measuring 25 mm or greater in the short axis, bone metastases with a soft tissue component 10 mm or greater in the longest diameter, solid organ metastases 10 mm or greater in the longest diameter, and all intraprostatic lesions had to be PSMA-positive on ^{68}Ga-PSMA-11 PET-CT. Read criteria differed from VISION only in the requirement for any intraprostatic lesions to be PSMA-positive in PSMAfore.[33,37] Full eligibility criteria are provided in the protocol (appendix 2 pp 67–70).

Randomisation and masking

Patients were randomly allocated (1:1) to the ^{177}Lu-PSMA-617 group or the ARPI change group with use of interactive response technology, and were stratified according to previous ARPI use (castration-resistant *vs* hormone-sensitive prostate cancer), and by pain symptoms (asymptomatic or mildly symptomatic [score 0–3] *vs* symptomatic [score >3] based on item 3 of the screening Brief Pain Inventory—Short Form [BPI-SF]). Before randomisation, abiraterone or enzalutamide was chosen by the investigator as the change of ARPI; the choice was disregarded if participants were randomly allocated to receive ^{177}Lu-PSMA-617. Treatment assignment was open to participants, investigator staff, personnel performing assessments, and the Novartis study team. However, no aggregate summaries by treatment group were done by Novartis until the primary radiographic progression-free survival analysis. Independent central reviewers were masked to randomised treatment assignment.

See Online for appendix 2

Procedures

Patients in the ^{177}Lu-PSMA-617 group received ^{177}Lu-PSMA-617 at a dosage of 7·4 GBq (200 mCi) ± 10% once every 6 weeks for six cycles. Those in the ARPI change group received either abiraterone or enzalutamide,

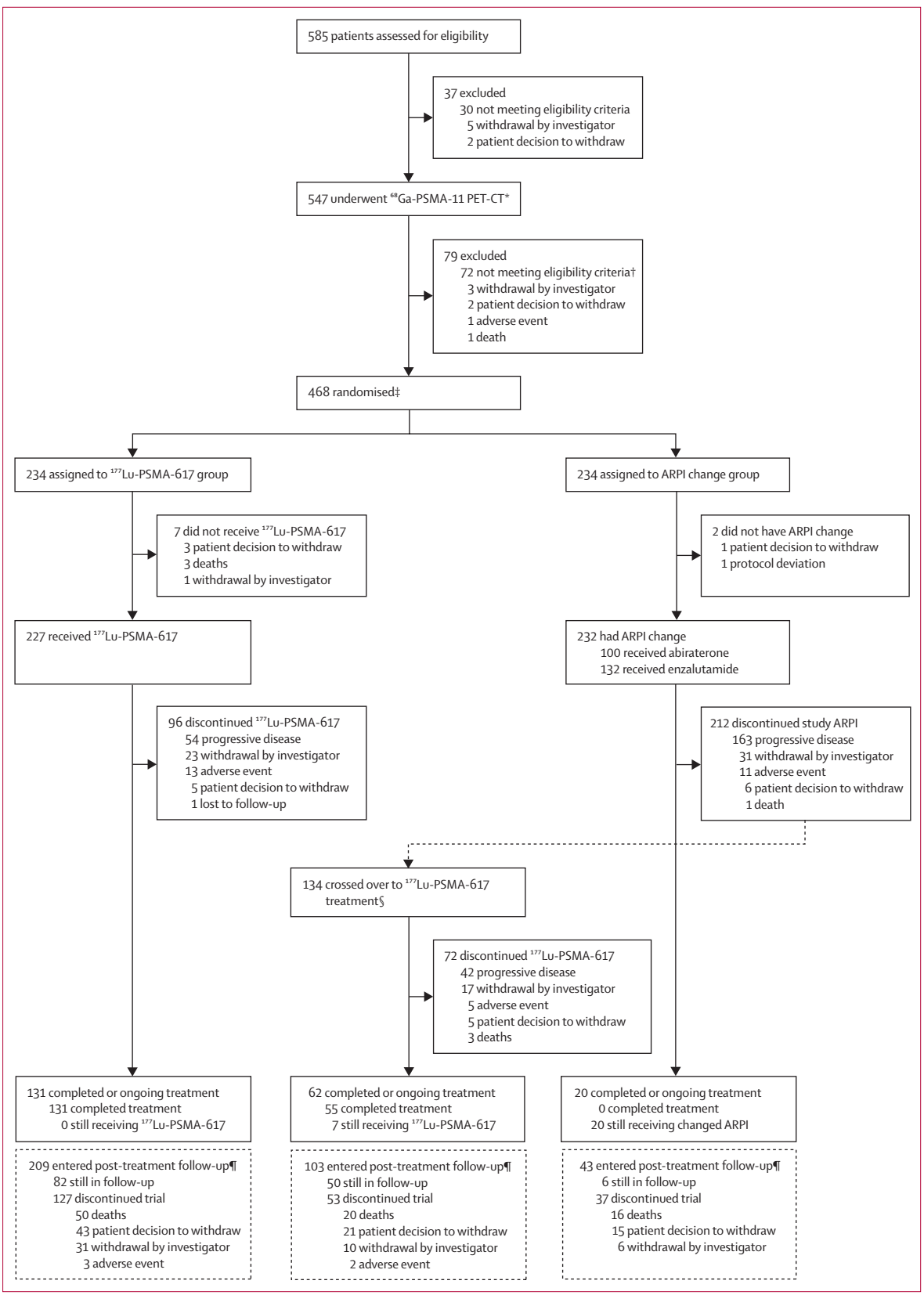

Figure 1: **Screening, randomisation, and follow-up at the time of third data cutoff**

ARPI=androgen receptor pathway inhibitor. PSMA=prostate-specific membrane antigen. *Status independent of trial enrolment: 505 met PSMA PET inclusion criteria (of the 518 who had one or more PSMA-positive lesions, 13 also had one or more exclusionary PSMA-negative lesions meeting size criteria), 24 did not meet PSMA PET inclusion criteria, and 18 had missing data. †24 did not meet PSMA PET inclusion criteria; 37 met PSMA PET imaging criteria but were excluded because they did not meet the other eligibility criteria, or for other reasons; and 18 had missing data. ‡One additional patient who underwent randomisation and was assigned to ^{177}Lu-PSMA-617 is not shown because they had a protocol deviation of informed consent not signed and were excluded from all analyses; one patient was randomised after the data cutoff for the primary analysis. §Patients eligible to cross over to ^{177}Lu-PSMA-617 had discontinued study ARPI treatment with radiographic disease progression confirmed by blinded independent central review per Response Evaluation Criteria in Solid Tumours v1.1. ¶Not including patients who were still receiving treatment at third data cutoff, or who never received treatment.

per the treating investigator's selection (appendix 1 p 6), and could cross over to ^{177}Lu-PSMA-617 upon meeting eligibility criteria (appendix 1 p 6), including radiographic disease progression. For this study, radiographic disease progression was defined according to the Prostate Cancer Working Group 3 (PCWG3)-modified Response Evaluation Criteria in Solid Tumours (RECIST) v1.1,[38] and confirmed by blinded independent central review. Crossover was not allowed upon clinical progression in the absence of centrally confirmed radiographic progression. Permitted and prohibited concomitant therapies are detailed in appendix 1 (p 6).

Treatment continued until completion of six cycles of ^{177}Lu-PSMA-617, unacceptable toxicity, radiographic progression, requirement for a prohibited therapy, or withdrawal. Participants had CT or MRI and ^{99m}Tc bone scans every 8 weeks for 24 weeks, then every 12 weeks, until radiographic disease progression. Participants attended follow-up visits within 1 week of their last treatment cycle or radiographic progression, 1 month thereafter for safety, then every 12 weeks (appendix 1 p 9).

Outcomes

The primary endpoint was radiographic progression-free survival, defined as the time from randomisation to radiographic disease progression (assessed by blinded independent central review per PCWG3-modified[38] RECIST v1.1), or death. The key secondary endpoint was overall survival, defined as time from randomisation to death from any cause. Additional prespecified secondary endpoints, including those not reported here, are listed and defined in appendix 1 (p 14). Those reported in this Article included progression-free survival (based on radiographic, clinical, or prostate-specific antigen [PSA] progression, as judged by investigator assessment); PSA response (defined as a ≥50% decrease in PSA concentration from baseline); time to symptomatic skeletal event; second-progression-free survival (based on investigator-assessed radiographic, clinical, or PSA progression on a subsequent line of therapy, including crossover therapy); health-related quality of life, based on patient-reported Functional Assessment of Cancer Therapy—Prostate (FACT-P) and BPI-SF (appendix 1 p 7); and safety and tolerability. The following secondary endpoints were immature at the time of the third interim overall survival analysis and are not reported here: second radiographic progression-free survival (for patients who crossed over), time to soft tissue progression, and time to chemotherapy. EQ-5D-5L data will be presented in a future report on health-related quality of life. The study is ongoing ahead of the final analysis of overall survival.

Prespecified exploratory endpoints are listed and defined in appendix 1 (p 14); objective response rate and time to PSA progression (based on an increase of ≥25% and ≥2 ng/mL from the nadir or from baseline beyond 12 weeks) are reported in this Article.

Efficacy was analysed in all randomly allocated participants (intention-to-treat population). Safety was assessed in all participants who received at least one dose of study treatment. Adverse events were assessed and graded according to the Common Terminology Criteria

	^{177}Lu-PSMA-617 group (n=234)	ARPI change group (n=234)
Demographic characteristics		
Region		
Europe	187 (80%)	182 (78%)
North America	47 (20%)	52 (22%)
Age, years		
Median (IQR)	71·0 (65·0–77·0)	72·0 (67·0–77·0)
<65	50 (21%)	44 (19%)
≥65	184 (79%)	190 (81%)
OMB category		
White	212 (91%)	214 (91%)
Black or African-American	7 (3%)	5 (2%)
Asian	1 (<1%)	2 (1%)
Unknown	14 (6%)	13 (6%)
Clinical characteristics		
ECOG performance status		
0	146 (62%)	116 (50%)
1	86 (37%)	114 (49%)
Missing	2 (1%)	4 (2%)
Site of disease		
Lung	21 (9%)	23 (10%)
Liver	13 (6%)	7 (3%)
Lymph node	76 (32%)	73 (31%)
Bone	205 (88%)	203 (87%)
RECIST v1.1 soft tissue lesions		
Target only	24 (10%)	25 (11%)
Non-target only	39 (17%)	39 (17%)
Target and non-target	58 (25%)	62 (26%)
None	113 (48%)	108 (46%)
Haemoglobin, g/L	128·0 (118·0–135·0), n=227	129·0 (122·0–137·0), n=232
Alkaline phosphatase, IU/L	100·0 (72·0–158·0), n=227	103·5 (80·0–146·0), n=232
Lactate dehydrogenase, IU/L	197·0 (169·0–234·0), n=223	196·5 (170·0–227·0), n=230
Prostate-specific antigen, µg/L	18·4 (7·0–51·7), n=230	14·9 (5·9–38·3), n=229
Prostate cancer history		
Time since diagnosis, years	5·45 (2·80–10·60)	6·80 (3·20–10·90)
Gleason score		
3	0	2 (1%)
4	1 (<1%)	3 (1%)
5	4 (2%)	1 (<1%)
6	15 (6%)	11 (5%)
7	69 (29%)	95 (41%)
8	46 (20%)	46 (20%)
9	76 (32%)	55 (24%)
10	15 (6%)	9 (4%)
Missing	8 (3%)	12 (5%)
Previous surgery (non-biopsy)	82 (35%)	96 (41%)
Previous radiotherapy	150 (64%)	172 (74%)

(Table 1 continues on next page)

	^{177}Lu-PSMA-617 group (n=234)	ARPI change group (n=234)
(Continued from previous page)		
Previous chemotherapy setting		
Neoadjuvant	1 (<1%)	0
Adjuvant	1 (<1%)	1 (<1%)
Metastatic	4 (2%)*	5 (2%)*
Previous anticancer regimens		
1	43 (18%)	46 (20%)
2	119 (51%)	118 (50%)
3	54 (23%)	53 (23%)
4	13 (6%)	13 (6%)
5	4 (2%)	3 (1%)
6	0	1 (<1%)
10	1 (<1%)	0
Setting at last anticancer regimen		
Adjuvant	2 (1%)	4 (2%)
Locally advanced or premetastatic	7 (3%)	10 (4%)
Metastatic	225 (96%)	220 (94%)
Previous ARPI		
Abiraterone	119 (51%)	130 (56%)†
Enzalutamide	94 (40%)	83 (35%)
Other‡	21 (9%)	21 (9%)
Previous ARPI setting§		
Castration-resistant prostate cancer	190 (81%)	190 (81%)
Hormone-sensitive prostate cancer	44 (19%)	44 (19%)
Pain symptoms§		
Asymptomatic or mildly symptomatic¶	165 (71%)	166 (71%)
Symptomatic‖	69 (29%)	68 (29%)

Data are n (%) or median (IQR) and are shown for all randomly allocated patients. Percentages might not total 100 because of rounding. ARPI=androgen receptor pathway inhibitor. BPI-SF=Brief Pain Inventory-Short Form. ECOG=Eastern Co-operative Oncology Group. OMB=Office of Management and Budget. PSMA=prostate-specific membrane antigen. RECIST=Response Evaluation Criteria in Solid Tumours. *Protocol deviations. †Two patients discontinued abiraterone because of toxicity and subsequently received darolutamide before randomisation. ‡Apalutamide or darolutamide. §Randomisation stratification factor. ¶Score of 0–3 on BPI-SF item 3 (worst pain intensity scale). ‖Score of >3 on BPI-SF item 3 (worst pain intensity scale).

Table 1: **Patient baseline characteristics**

for Adverse Events v5.0 and Medical Dictionary for Regulatory Activities v26.0.

Statistical analysis

The primary analysis of radiographic progression-free survival was prespecified to occur after approximately 156 events in 450 patients (first data cutoff), providing 95% power to detect a hazard ratio (HR) of 0·56 at an overall one-sided α of 0·025 (appendix 3 p 31). The key secondary endpoint was planned to be tested if the primary endpoint was met in the primary analysis. A Lan–DeMets (O'Brien–Fleming) α spending function was used to control type I error across four-look testing of overall survival (appendix 3 pp 62–63). Interim analyses of overall survival were prespecified. The primary intent of the interim analyses was to stop early for superior efficacy; there was no intent to assess futility at the interim analyses. The first interim overall survival analysis was planned to occur at the time of the primary analysis (first data cutoff) of radiographic progression-free survival. Second and third interim overall survival analyses were planned to occur at approximately 125 (second data cutoff) and 223 (third data cutoff) deaths, respectively. Final analysis of overall survival is planned to occur after approximately 297 deaths (final data cutoff), providing 80% power to detect an HR of 0·72 (not considering the impact of crossover) at an overall one-sided α of 0·025. Updated analyses of radiographic progression-free survival could occur as exploratory analyses at the time of the second and third data cutoffs. In this primary report of PSMAfore, we present the primary (first data cutoff) and updated (third data cutoff) analyses of radiographic progression-free survival; all other endpoints are based on the third data cutoff.

See Online for appendix 3

Time-to-event endpoints were assessed using the log-rank test stratified by the randomisation factors. HRs and 95% CIs were estimated using stratified Cox proportional hazards models. Medians and 95% CIs were estimated using the Kaplan–Meier method. p values were converted to two-sided for the primary endpoint. Analysis methods for other endpoints and further details are provided in appendix 1 (pp 7–8).

Role of the funding source

In collaboration with the authors, the funder of the study, Novartis, was involved in study design, data collection, data analysis, data interpretation, and the writing and review of the report.

Results

Of 585 patients screened, 547 underwent ^{68}Ga-PSMA-11 PET-CT, of whom 505 (92%) met the PSMA PET inclusion criteria, independent of trial enrolment (figure 1). Overall, 468 (80%) of the 585 patients screened met all the eligibility criteria and were randomly allocated between June 15, 2021 and Oct 7, 2022 to receive either ^{177}Lu-PSMA-617 (n=234) or ARPI change (n=234). Protocol deviations are listed in appendix 1 (p 15).

The primary analysis (first data cutoff Oct 2, 2022), second interim overall survival analysis (second data cutoff June 21, 2023), and third interim overall survival analysis (third data cutoff Feb 27, 2024) were complete at the time of publication of this Article. All data presented in this Article, other than the primary analysis of radiographic progression-free survival (first data cutoff), are based on the third data cutoff. Data corresponding to the second data cutoff have been previously presented.[39] The median time from randomisation to first data cutoff was 7·26 months (IQR 3·38–10·55) and to third data cutoff was 24·11 months (20·24–27·40). At the third data cutoff, 134 (57%) of 234 patients allocated to the ARPI change group had crossed over to receive ^{177}Lu-PSMA-617, representing 78% of patients who had radiographic progression confirmed by blinded independent central review.

Baseline patient characteristics were generally well balanced between groups (table 1). Abiraterone was the previous ARPI received in 119 (51%) of 234 patients in the ^{177}Lu-PSMA-617 group and in 130 (56%) of 234 patients in the ARPI change group. The proportion of patients with an ECOG performance status of 0 was higher in the ^{177}Lu-PSMA-617 group (146 [62%]) than in the ARPI change group (116 [50%]).

After discontinuation of study treatment, the most common postprotocol anticancer therapies received were chemotherapy (110 [47%] of 234 patients in the ^{177}Lu-PSMA-617 group and 99 [42%] of 234 in the ARPI change group), radiotherapy (44 [19%] and 41 [18%], respectively), and hormonal therapy (31 [13%] and five [2%], respectively). Other postprotocol anticancer therapies were received by less than 5% of patients in either group (appendix 1 p 15).

In the primary analysis (first data cutoff) after 166 radiographic progression-free survival events, the HR for the primary efficacy endpoint of radiographic progression or death in the ^{177}Lu-PSMA-617 group versus the ARPI change group was 0·41 (95% CI 0·29–0·56; two-sided p<0·0001); median radiographic progression-free survival time was 9·30 months (95% CI 6·77–not estimable [NE]) in the ^{177}Lu-PSMA-617 group versus 5·55 months (4·04–5·95) in the ARPI change group (appendix 1 p 10). The median follow-up time for radiographic progression-free survival in all randomly allocated patients was 3·71 months (IQR 2·14–6·05) in the ^{177}Lu-PSMA-617 group (median

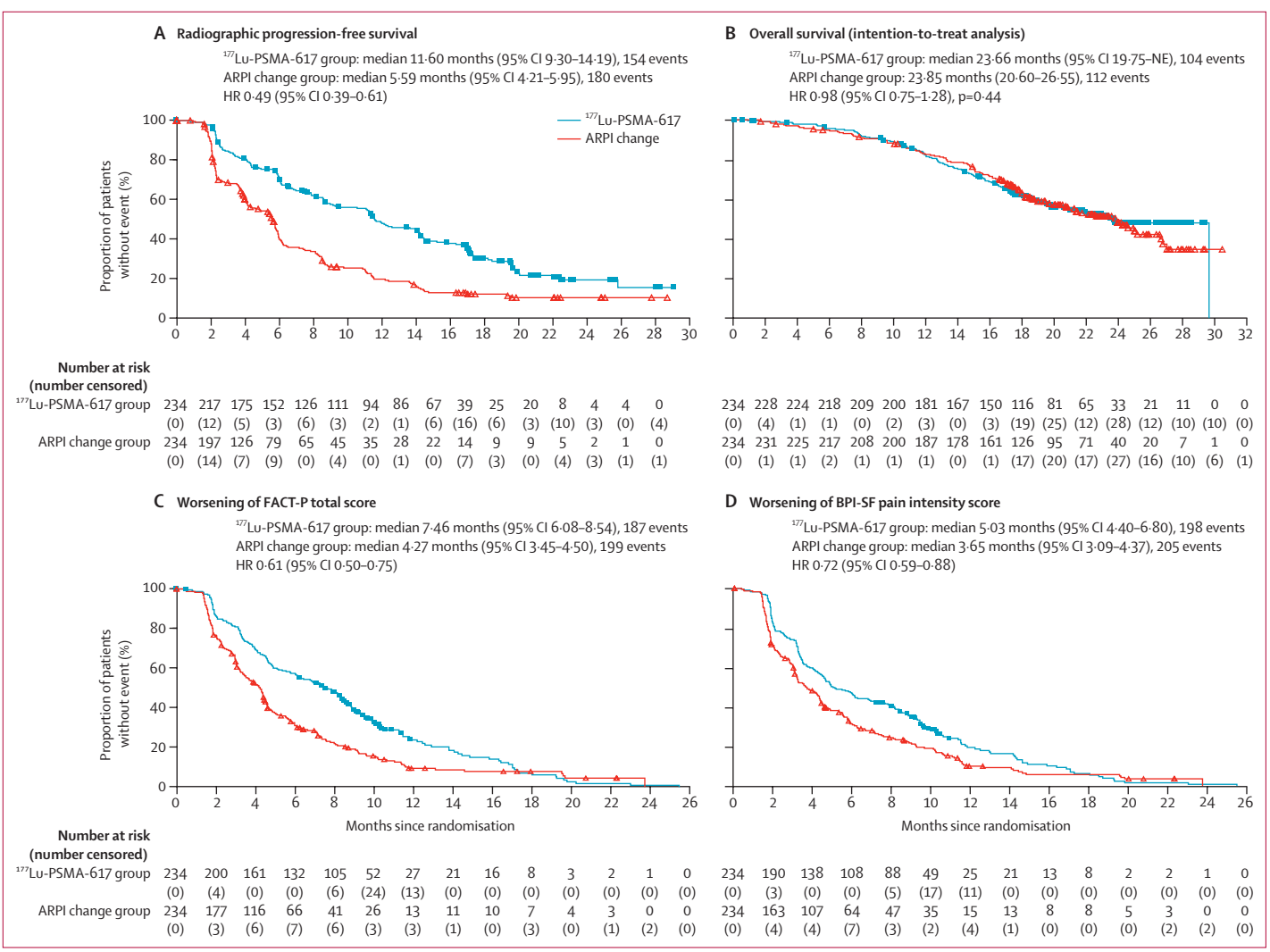

Figure 2: **Time-to-event endpoints at the time of third data cutoff**

(A) Updated radiographic progression-free survival (see appendix 1 p 10 for primary analysis); events were radiographic disease progression (determined by blinded independent central review per the Prostate Cancer Clinical Trials Working Group 3-modified[38] Response Evaluation Criteria in Solid Tumours v1.1) or death. (B) Overall survival (intention-to-treat analysis; three patients died before receiving ^{177}Lu-PSMA-617). (C) Time to worsening of FACT-P total score; events were a decrease of ≥10 points,[40,41] clinical disease progression, or death. (D) Time to worsening on BPI-SF pain intensity scale; events were an increase of ≥2 points,[42,43] clinical disease progression, or death. See appendix 1 (p 9) for the schedule of investigations. ARPI=androgen receptor pathway inhibitor. BPI-SF=Brief Pain Inventory—Short Form. FACT-P=Functional Assessment of Cancer Therapy—Prostate. HR=hazard ratio. NE=not estimable. PSMA=prostate-specific membrane antigen.

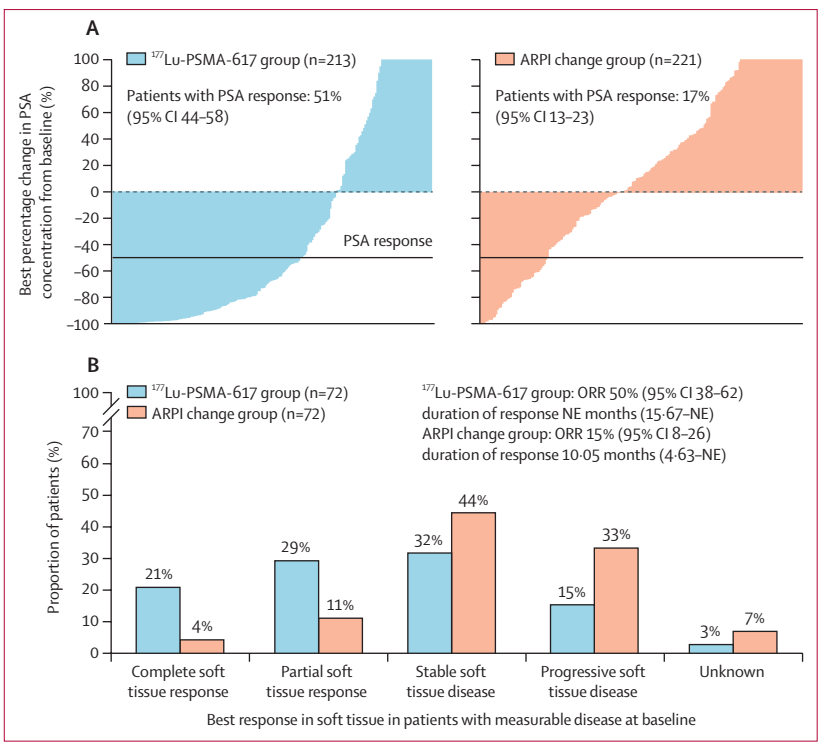

Figure 3: **PSA responses and soft tissue responses at the time of third data cutoff**
(A) Changes in PSA concentration from baseline; PSA response was defined as a confirmed decrease of 50% or greater; secondary endpoint. Data are from patients with available PSA measurements at the time of the third data cutoff; assessments were at 3, 6, and 12 months after randomisation. Increases greater than 100% are truncated to 100%. (B) Soft tissue responses (based on RECIST v1.1, excluding bone lesion progression; exploratory endpoint). Data are from a subset of patients with measurable disease at baseline. ORR is the sum of complete soft tissue response and partial soft tissue response. Median duration of response (95% CI) was estimated using the Kaplan–Meier method in patients with an objective response, and defined as the duration from first documented best overall response to documented disease progression (determined by blinded independent central review per Prostate Cancer Clinical Trials Working Group 3-modified[38] RECIST v1.1) or death. ARPI=androgen receptor pathway inhibitor. NE=not estimable. ORR=objective response rate. PSA=prostate-specific antigen. PSMA=prostate-specific membrane antigen. RECIST= Response Evaluation Criteria in Solid Tumours.

number of cycles 3·0 [IQR 2·0–5·0] among patients who received at least one dose of ^{177}Lu-PSMA-617; appendix 1 p 16) and 2·35 months (1·81–5·55) in the ARPI change group.

At the third data cutoff after 334 radiographic progression-free survival events, the updated HR for radiographic progression or death in the ^{177}Lu-PSMA-617 group versus the ARPI change group was 0·49 (95% CI 0·39–0·61); median radiographic progression-free survival time was 11·60 months (95% CI 9·30–14·19) in the ^{177}Lu-PSMA-617 group versus 5·59 months (4·21–5·95) in the ARPI change group (figure 2A). The median follow-up time for radiographic progression-free survival in all randomly allocated patients was 8·77 months (IQR 3·98–17·05) in the ^{177}Lu-PSMA-617 group (median number of cycles 6·0 [IQR 4·0–6·0] among patients who received at least one dose of ^{177}Lu-PSMA-617; appendix 1 p 16) and 4·27 months (2·14–8·51) in the ARPI change group. In prespecified subgroup analyses of radiographic progression-free survival, HRs were less than 1 for ^{177}Lu-PSMA-617 versus ARPI change in all subgroups (appendix 1 p 11).

At the third data cutoff, there were 216 deaths, corresponding to 73% of the number of deaths prespecified in the statistical analysis plan (appendix 3 p 63) for the final analysis of overall survival, the key secondary endpoint. Death occurred in 104 (44%) of 234 patients in the ^{177}Lu-PSMA-617 group, including three deaths before treatment was started. Death occurred in 112 (48%) of 234 patients in the ARPI change group, including in 59 (44%) of 134 patients who crossed over to ^{177}Lu-PSMA-617 treatment. Most deaths were related to prostate cancer in both the ^{177}Lu-PSMA-617 group (81 [78%] of 104) and the ARPI change group (94 [84%] of 112). In the intention-to-treat analysis, the HR for overall survival in the ^{177}Lu-PSMA-617 group versus the ARPI change group was 0·98 (95% CI 0·75–1·28; p=0·44); median overall survival time was 23·66 months (95% CI 19·75–NE) in the ^{177}Lu-PSMA-617 group versus 23·85 months (20·60–26·55) in the ARPI change group (figure 2B). The median follow-up time for overall survival in all randomly allocated patients was 17·94 months (IQR 12·72–22·60) in the ^{177}Lu-PSMA-617 group and 18·45 months (14·59–22·80) in the ARPI change group.

Among other secondary endpoints, at the third data cutoff, the proportion of patients with a confirmed PSA response was higher in the ^{177}Lu-PSMA-617 group (110 of 217; 51% [95% CI 44–58]) than in the ARPI change group (39 of 225; 17% [13–23]; figure 3A). Additionally, the ^{177}Lu-PSMA-617 group had a longer time to first symptomatic skeletal event (median NE [95% CI NE–NE] *vs* 17·97 months [14·26–NE]; HR 0·41 [95% CI 0·26–0·63]; appendix 1 p 17), longer progression-free survival (6·70 months [95% CI 5·49–8·77] *vs* 3·12 months [3·02–3·29]; HR 0·43 [0·35–0·52]; appendix 1 p 12), and longer second-progression-free survival (18·00 months [15·70–20·53] *vs* 15·31 months [13·37–20·21]; HR 0·86 [0·67–1·11]; appendix 1 p 13) than the ARPI change group.

At the third data cutoff, median time to worsening in FACT-P total score (as a secondary, health-related quality-of-life endpoint) was longer in the ^{177}Lu-PSMA-617 group (7·46 months [95% CI 6·08–8·54]) than in the ARPI change group (4·27 months [3·45–4·50]; HR 0·61 [0·50–0·75]; figure 2C), as was time to worsening in BPI-SF pain intensity score (figure 2D).

Among the exploratory endpoints, the objective response rate for soft tissue disease, excluding bone lesion progression, in patients with measurable disease at baseline was higher in the ^{177}Lu-PSMA-617 group (36 of 72; 50% [95% CI 38–62]) than in the ARPI change group (11 of 72; 15% [8–26]), including 15 (21%) of 72 patients versus three (4%) of 72 who had complete soft tissue response (figure 3B; appendix 1 pp 18–19).

Median time to PSA progression was longer in the ^{177}Lu-PSMA-617 group (10·64 months [95% CI 8·57–12·68]) than in the ARPI change group

(4·24 months [3·48–4·44]; HR 0·40 [95% CI 0·31–0·50]; appendix 1 p 12).

At the third data cutoff, patients in the ^{177}Lu-PSMA-617 group had received a median cumulative dose of 42·39 GBq (IQR 29·27–43·54), and the median duration of exposure was 8·41 months (IQR 6·21–9·17; appendix 1 p 16). In the ARPI change group, the median duration of exposure was 6·52 months (3·29–11·06); 100 patients received abiraterone and 132 patients received enzalutamide.

The incidence of treatment-emergent adverse events of grade 3 or higher was lower in the ^{177}Lu-PSMA-617 group (81 [36%] of 227 patients in the safety population) than in the ARPI change group (112 [48%] of 232; table 2). These treatment-emergent adverse events were considered treatment-related by the investigator in 27 (12%) patients in the ^{177}Lu-PSMA-617 group and in 29 (13%) in the ARPI change group (appendix 1 p 20). Dry mouth, asthenia, nausea, and anaemia were the four most common treatment-emergent adverse events in the ^{177}Lu-PSMA-617 group. Most of these events were grade 1 or 2 (table 2). Thrombocytopenia was observed in 17 (7%) patients in the ^{177}Lu-PSMA-617 group and seven (3%) in the ARPI change group (appendix 1 p 20). Treatment-emergent adverse events leading to dose adjustment and treatment interruption were less frequent in the ^{177}Lu-PSMA-617 group than in the ARPI change group (table 2). The proportions of patients who had treatment-emergent adverse events leading to treatment discontinuation or death were low and similar between groups (table 2). There were no treatment-related adverse events that led to death in the ^{177}Lu-PSMA-617 group and one in the ARPI change group (cerebrovascular accident; appendix 1 p 20). Groupings of treatment-emergent adverse events according to safety topics of interest are shown in appendix 1 (p 21). The number of patients who had renal toxicity adverse events was similar in the ^{177}Lu-PSMA-617 group (15 [7%]) and the ARPI change group (18 [8%]; appendix 1 p 21). Serious treatment-related adverse events occurred in six (3%) patients in the ^{177}Lu-PSMA-617 group and six (3%) in the ARPI change group (appendix 1 p 20). The serious treatment-related adverse events in the ^{177}Lu-PSMA-617 group were grade 3 or higher anaemia in two (1%) patients, thrombocytopenia in one (<1%), dysphagia in one (<1%), increased alanine aminotransferase in one (<1%), increased aspartate aminotransferase in one (<1%), and sciatica in one (<1%); and grade 1–2 anaemia in one patient (<1%). In the ARPI change group, the serious treatment-related adverse events were all of grade 3 or higher: anaphylactic reaction in one (<1%) patient, pneumonia in one (<1%), hypercalcaemia in one (<1%), pathological fracture in one (<1%), hypertension in one (<1%), and cerebrovascular accident in one (<1%). Adverse events relating to ^{68}Ga-PSMA-11 are detailed in appendix 1 (p 22).

	^{177}Lu-PSMA-617 group (n=227)		ARPI change group (n=232)	
	Any grade	Grade ≥3	Any grade	Grade ≥3
Any	224 (99%)	81 (36%)*	226 (97%)	112 (48%)*
Occurring in >10% of patients				
Dry mouth	131 (58%)†	2 (1%)	6 (3%)†	0
Asthenia	74 (33%)	2 (1%)	67 (29%)	8 (3%)
Nausea	72 (32%)	0	27 (12%)	1 (<1%)
Anaemia	61 (27%)	14 (6%)	44 (19%)	16 (7%)
Fatigue	53 (23%)	1 (<1%)	59 (25%)	4 (2%)
Constipation	50 (22%)	1 (<1%)	33 (14%)	0
Decreased appetite	49 (22%)	0	43 (19%)	1 (<1%)
Arthralgia	45 (20%)	0	54 (23%)	1 (<1%)
Diarrhoea	38 (17%)	0	21 (9%)	1 (<1%)
COVID-19	36 (16%)	1 (<1%)	27 (12%)	1 (<1%)
Back pain	31 (14%)	3 (1%)	46 (20%)	6 (3%)
Vomiting	26 (11%)	0	11 (5%)	0
Oedema peripheral	19 (8%)	0	28 (12%)	0
Pain in extremity	17 (7%)	1 (<1%)	25 (11%)	1 (<1%)
Weight loss	15 (7%)	1 (<1%)	32 (14%)	3 (1%)
Leading to adjustment of dose	8 (4%)	3 (1%)	36 (16%)	7 (3%)
Leading to interruption of treatment	28 (12%)	13 (6%)	45 (19%)	20 (9%)
Leading to discontinuation of treatment	13 (6%)	8 (4%)	12 (5%)	9 (4%)
Leading to death	4 (2%)	4 (2%)	5 (2%)	5 (2%)
Cardiac arrest	1 (<1%)‡	1 (<1%)‡	1 (<1%)‡	1 (<1%)‡
Cerebrovascular accident	0	0	1 (<1%)§	1 (<1%)§
Coma	0	0	1 (<1%)‡	1 (<1%)‡
COVID-19 pneumonia	1 (<1%)‡	1 (<1%)‡	0	0
Dyspnoea	0	0	1 (<1%)‡	1 (<1%)‡
Intestinal ischaemia	1 (<1%)‡	1 (<1%)‡	0	0
Multiple organ dysfunction syndrome	0	0	1 (<1%)‡	1 (<1%)‡
Sepsis	1 (<1%)‡	1 (<1%)‡	0	0

Data are n (%) of patients with an event, and are shown for patients who were randomly allocated and received at least one dose of their assigned treatment; seven patients who were randomly allocated to the ^{177}Lu-PSMA-617 group did not receive treatment. ARPI=androgen receptor pathway inhibitor. PSMA=prostate-specific membrane antigen. *Grade 3–4: 80 (35%) patients in the ^{177}Lu-PSMA-617 group; 111 (48%) patients in the ARPI change group; these numbers include some patients who also had a grade 5 event. †Grade 2 dry mouth: 19 (8%) patients in ^{177}Lu-PSMA-617 group; 0 patients in ARPI change group. ‡Considered not related to study drug by the investigator. §Considered related to study drug by the investigator.

Table 2: **Overview of treatment-emergent adverse events at the time of third data cutoff**

Discussion

Taxane-naive patients with metastatic castration-resistant prostate cancer who have progressed on a first ARPI face an important decision on the order of subsequent therapies. Taxane-based chemotherapy has proven efficacious in relieving symptoms, delaying disease progression, and prolonging survival, but its toxicity can negatively affect quality of life during treatment.[8,13,14] Many patients forego taxanes in this clinical context in favour of switching to an alternative ARPI.[2–5,12] This trial shows that ^{177}Lu-PSMA-617 offers a more effective therapy option for controlling disease and maintaining quality of life than changing ARPI. The results of this study are complementary to those

of the pivotal VISION study, which assessed ^{177}Lu-PSMA-617 in a post-taxane patient population.[33]

Treatment with ^{177}Lu-PSMA-617 led to a clinically meaningful prolongation of radiographic progression-free survival compared with ARPI change in taxane-naive patients with PSMA-positive metastatic castration-resistant prostate cancer. ^{177}Lu-PSMA-617 had a favourable safety profile, with a lower incidence of grade 3–4 or serious adverse events despite a longer median duration of exposure than the study ARPI treatment. Other patient-relevant and clinically important endpoints also favoured ^{177}Lu-PSMA-617 over ARPI change, including time to symptomatic skeletal events, health-related quality of life, and radiographic and biochemical (PSA) response rates, as well as investigator-assessed progression-free survival and second-progression-free survival on the next line of therapy.

The estimation of median radiographic progression-free survival was immature at the time of the primary analysis owing to the short follow-up; median radiographic progression-free survival had matured by the third data cutoff, showing a clinically meaningful difference between medians in favour of ^{177}Lu-PSMA-617.

Overall survival was the key secondary endpoint in this study. At the time of the third data cutoff, there was no difference in overall survival between the ^{177}Lu-PSMA-617 and ARPI change groups in the intention-to-treat analysis. The trial was designed to allow crossover from ARPI change to ^{177}Lu-PSMA-617 to offer access to ^{177}Lu-PSMA-617 in a patient-centric study design. The high crossover rate (134 [57%] of 234 patients assigned to ARPI change) might have confounded the analysis of overall survival. Methods for adjusting overall survival for the effect of crossover will be explored in the planned final overall survival analysis.

PSMAfore compared ^{177}Lu-PSMA-617 with a change of ARPI in patients who, in conjunction with their consulting physician, felt that it was appropriate to change ARPI and delay taxane-based chemotherapy. A change of ARPI is clinically less effective than taxanes,[5] and it might not be suitable for all patients. However, real-world data indicate that many patients do not receive taxane-based chemotherapy after progression on a first ARPI and instead switch to a second ARPI.[2–5,12] In this study, median radiographic progression-free survival in the ARPI change group was 5·59 months at the third data cutoff, which is longer than previously reported in several studies,[23,44,45] and PSA response was consistent with published literature,[21,23,46] supporting the use of ARPI change as a control treatment for the PSMAfore population. Only patients who were candidates for a change of ARPI were enrolled, and this was reflected in the baseline patient characteristics, which were consistent with patients having more indolent disease. For example, the proportion of patients with liver metastases in this trial was low (6% in the ^{177}Lu-PSMA-617 group and 3% in the ARPI change group), and was consistent with a phase 2 trial of enzalutamide and abiraterone sequencing in patients with newly diagnosed metastatic castration-resistant prostate cancer.[19] Patients could discontinue the trial at any time before or after randomisation to receive other anticancer therapies, including chemotherapy; patients with clinical progression had the option to discontinue randomised treatment and receive chemotherapy, even in the absence of radiographic progression. At the third data cutoff, less than half of patients had received subsequent chemotherapy. These observations combined with the multiple patient-centric benefits afforded with ^{177}Lu-PSMA-617 suggest that a change in drug class to ^{177}Lu-PSMA-617 could offer an acceptable alternative to change of ARPI for second-line therapy in metastatic castration-resistant prostate cancer.[2,18–21] PSMAfore did not have a bidirectional crossover design, so conclusions cannot be made about the optimal sequencing of change of ARPI versus second-line ^{177}Lu-PSMA-617.

PSMA PET imaging criteria were met in 92% of patients who underwent ^{68}Ga-PSMA-11 PET-CT scanning, while 87% of patients met the PSMA PET imaging criteria in VISION.[33] Further analyses of PSMA PET imaging parameters are planned.

This study has several limitations. First, it did not examine a population with more aggressive disease for whom taxane-based chemotherapy was the most appropriate next therapy after progression on an ARPI; therefore, the results cannot be generalised to this population. Phase 2 trials of ^{177}Lu-PSMA-617 in the pre-taxane metastatic castration-resistant prostate cancer setting versus docetaxel (NCT04663997) and in the metastatic hormone-sensitive prostate cancer setting (NCT04343885) before docetaxel are ongoing. However, future larger studies using a chemotherapy comparator are warranted for this population at increased risk. Second, PSMAfore did not assess the differential efficacy of ^{177}Lu-PSMA-617 in patients with or without pathogenic HRR gene alterations, which are associated with high tumour PSMA expression[47] and are recognised in metastatic castration-resistant prostate cancer.[48–51] Patients with known HRR gene alterations were ineligible. However, genotyping was not required during screening, so it is not known what proportion of patients had HRR mutations, and the results cannot be generalised to patients with HRR gene alterations. Finally, double blinding was not possible in this trial because of the different administration routes of the trial interventions and because of radiation protection regulations and ethical considerations. This could have biased the reporting of clinical and safety endpoints, particularly those which have an element of subjectivity, including health-related quality of life and adverse event reporting. The lack of double blinding might also have influenced the ongoing management of patients.

In conclusion, PSMAfore met the primary endpoint, showing significant prolongation of time to radiographic

progression or death with ^{177}Lu-PSMA-617 compared with ARPI change in taxane-naive patients with metastatic castration-resistant prostate cancer who had disease progression on a previous ARPI. In addition, ^{177}Lu-PSMA-617 was more effective than a change of ARPI in prolonging the time to PSA progression, symptomatic skeletal events, deterioration in health-related quality of life, and worsening of pain, with a favourable safety profile. Overall survival was similar with ^{177}Lu-PSMA-617 and ARPI change; whether or not the high rate of crossover confounded the analysis of overall survival will be explored at the time of the final analysis. These results support ^{177}Lu-PSMA-617 as a treatment option for patients with metastatic castration-resistant prostate cancer who have initially progressed on an ARPI and who are being considered for a change of ARPI as their next therapy.

Contributors

MJM, KH, JSdB, NDS, KNC, MC, OS, and KF contributed to conceptualisation. EK and SG contributed to data curation and analysis. MJM, DC, KH, JSdB, NDS, KNC, JMP, AF, XXW, HM, GR, HŠ, JN, BM, and ÁM-P contributed to investigation. TNK, CW, KL, EK, and SG contributed to trial oversight. MJM, EK, SG, TNK, CW, and KL accessed and verified the underlying data in the study. All authors contributed to data interpretation and writing and reviewing the manuscript, and accept responsibility for the decision to submit for publication.

Declaration of interests

MJM declares receiving consulting or advisory fees from Amgen, AstraZeneca, Blue Earth Diagnostics, Clarity Pharmaceuticals, Convergent Therapeutics, Daiichi, ITM Isotope Technologies Munich, Lantheus Medical Imaging, Pfizer, POINT Biopharma, Progenics, Telix Pharmaceuticals, and Z-Alpha; stock and other ownership in Doximity; travel, accommodations, and expenses from APCCC, AstraZeneca, and Memorial Sloan-Kettering Cancer Center; institutional research funding from Astellas Pharma, Bayer, Celgene, Corcept Therapeutics, Janssen, Novartis, Progenics, and Roche/Genentech; patents pending with Novartis; and royalties from Telix. DC declares receiving payment or honorarium from Astellas, AstraZeneca, Bayer, Bristol Myers Squibb, Clovis, Eisai, Exelisis, GlaxoSmithKline, Ipsen, Janssen, Lilly, MSD, Pfizer, QED Therapeutics, and Roche; travel or other expenses from Astellas, AstraZeneca, Bristol Myers Squibb, GlaxoSmithKline, Ipsen, Janssen, MSD, Novartis, Pfizer, and Roche; participation on a data safety monitoring or advisory board for Astellas, AstraZeneca, Bristol Myers Squibb, GlaxoSmithKline, Ipsen, Janssen, MSD, Novartis, Pfizer, and Roche; and other financial or non-financial interests in Spanish Oncology Genito-Urinary Group Foundation and GUARD Consortium Spanish group. KH declares receiving grants or contracts from Boston Scientific, Janssen, and Novartis; consulting fees from Amgen, AstraZeneca, Bain Capital, Bayer, Boston Scientific, Convergent, Curium, Debiopharm, EcoR1, Fusion, GE Healthcare, Immedica, Isotopen Technologien München, Janssen, Merck, Molecular Partners, Novartis, NVision, POINT Biopharma, Pfizer, Radiopharm Theranostics, Rhine Pharma, Siemens Healthineers, Sofie Biosciences, Telix, Theragnostics, and Y-mAbs Therapeutics; and stock or other ownership in AdvanCell, Aktis Oncology, Convergent, NVision, Pharma 15, and Sofie Biosciences. JSdB declares receiving grants or contracts from Amgen, Astellas, AstraZeneca, Bayer, Bioxcel Therapeutics, Crescendo, Daiichi-Sankyo, Endocyte, Genentech/Roche, GlaxoSmithKline, ImCheck Therapeutics, Janssen, Merck Serono, Merck Sharp & Dohme, Novartis, Pfizer, and Sanofi Aventis; consulting or advisory fees from Astellas, AstraZeneca, Bayer, Daiichi-Sankyo, Genentech/Roche, GlaxoSmithKline, Janssen, Merck Serono, Merck Sharp & Dohme, Orion, Pfizer, Sanofi Aventis, and Taiho; payment or honorarium from Astellas, AstraZeneca, Bayer, Cellcentric, Crecendo, Daiichi, Genentech, Genmab, GlaxoSmithKline, Janssen, Merck Serono, Mycrix, MSD, Orion, Pfizer, Sanofi Aventis, and Taiho; being an inventor on patent 8822438 for a method for treating cancer; participating on a data safety monitoring or advisory board for Amgen, AstraZeneca, Bayer, Bioxcel Therapeutics, Crescendo, Daiichi-Sankyo, Endocyte, Genentech/Roche, GlaxoSmithKline, ImCheck Therapeutics, Janssen, Merck Serono, Merck Sharp & Dohme, Novartis, Oncternal, Pfizer, and Sanofi Aventis; and receiving institutional royalties or licences related to abiraterone, PARP inhibitor, and PI3K/AKT. NDS declares receiving support for the present manuscript from Novartis; consulting fees from AbbVie, Accord, Amgen, Antev, Arquer, Asieris, Astellas, AstraZeneca, Aura Biosciences, Bayer, Bioprotect, Bristol Myers Squibb, Clarity, Cold Genesys, Curium, Dendreon, Exact Imaging, Ferring, Fize Medical, Invitae, Janssen, Lantheus, Lilly, MDXHealth, Merck, Minomic, Myriad, Novartis, PlatformQ, Pfizer, POINT Biopharma, Preview, Promaxo, Propella, Protara, Sanofi Genzyme, Siemens, Speciality Networks, Sumitomo, Telix, Tolmar, and Urogen; having a leadership or fiduciary role for Photocure, and Alessa; and having stock or stock options with Photocure, and Alessa. KNC declares receiving grants or contracts from AstraZeneca, Bayer, Janssen, Merck, Novartis, Pfizer, POINT Biopharma, and Roche; and consulting fees from Astellas, AstraZeneca, Bayer, Janssen, Merck, Novartis, Pfizer, POINT Biopharma, and Roche. MC declares receiving support for the present manuscript from Novartis; consulting fees from Astellas, AstraZeneca, Blue Earth Diagnostics, Boston Scientific, Elekta, Johnson & Johnson, Lantheus, Macrogenics, Pfizer, Profound Medical, Sumitomo, Telix, Tolmar, and Tempus; payment or honorarium from Johnson & Johnson, Pfizer, Profound Medical, and Telix; participating on an advisory board for Telix; and having a leadership or fiduciary role for the Society of Nuclear Medical and Molecular Imaging, and the American Urological Association. JMP declares receiving grants or contracts from BeiGene, Bristol Myers Squibb, Immunocore, Mirati, MSD, and Pfizer; consulting fees from Astellas, AstraZeneca, BeiGene, Bristol Myers Squibb, Clovis, Ideaya, Immunocore, Janssen, MSD, Novartis, and Pfizer; and travel expenses from AstraZeneca, Janssen, and Pfizer. AF declares receiving payment or honorarium from Astellas, AstraZeneca, Bayer, Bristol Myers Squibb, Janssen, Merck, MSD, Novartis, and Pfizer; and travel expenses from Astellas, AstraZeneca, Bayer, Bristol Myers Squibb, Janssen, Merck, SD, Novartis, and Pfizer. XXW declares receiving institutional grants or contracts from Bristol Myers Squibb; consulting fees from Dendreon and Novartis; honorarium from Novartis; travel or other expenses from Novartis; and participating on a data safety monitoring or advisory board for Dendreon and Novartis. HM declares receiving payment or honorarium from Astellas, AstraZeneca, Bayer, Bristol Myers Squibb, Ipsen, Janssen, Merck, MSD, Novartis and Pfizer; travel or other expenses from AstraZeneca, Novartis, Pfizer, Roche; and acting on a scientific steering committee for Curium. GR declares receiving grants or contracts from Bayer; and consulting fees from Astellas, AstraZeneca, Bayer, Janssen, MSD, Novartis, and Pfizer. HŠ declares receiving consulting fees from Astellas, Bayer, Janssen, Novartis, and Pfizer. JN declares receiving contracts or grants from ABX, and Novartis; consulting fees from Curium, and POINT Biopharma; and payment or honorarium from Bayer AG, and Pfizer. BM declares receiving grants or contracts from Astellas, Bayer, Janssen, Roche, and Sanofi; payment or honorarium from Astellas, Bayer, Bristol Myers Squibb, Ipsen, Janssen, Pfizer, Roche, and Sanofi; travel or other expenses from Ipsen, Janssen Pfizer, and Roche; and other financial or non-financial interests in Astellas, Bayer, Bristol Myers Squibb, Ipsen, Janssen, Pfizer, Roche, and Sanofi. ÁM-P declares receiving personal fees from Astellas, AstraZeneca, Bayer, Bristol Myers Squibb, Ipsen, Janssen, Merck, MSD, Novartis, and Pfizer; and non-financial support from Bayer, Ipsen, Merck, and Pfizer. EK, SG, TNK, CW, and KL declare being employed by and receiving restricted stock options from Novartis. OS declares receiving support for the present manuscript from Novartis; institutional grants or contracts from Amgen, AstraZeneca, Bayer, Endocyte, Invitae, Janssen, Lantheus, Merck, Novartis, Progenics, and Tenebio; consulting fees from ARTBIO, AstraZeneca, Bayer, Blue Earth Diagnostics, Clarity Pharmaceuticals, Fusion, Isotopen Technologien Muenchen, Janssen, Merck, Myovant, Myriad, Noria Therapeutics, NorthStar, Novartis, Pfizer, POINT Biopharma, Sanofi, Telix, and Tenebio; travel and accommodation expenses from Lantheus, NorthStar, and Novartis; participation on a data safety monitoring or advisory board for AstraZeneca, Merck, and Pfizer; and stock or stock options in

ARTBIO, Clarity Pharmaceuticals, Convergent, Fusion, Lilly, Pfizer, Ratio, and Telix. KF declares receiving institutional honorarium or payments to Gustave Roussy Institute from Astellas, AstraZeneca, Bayer, Janssen, MSD, Novartis, and Sanofi; institutional participation on data safety monitoring or advisory boards for Amgen, Astellas, AstraZeneca, Bayer, Clovis, Daiichi Sankyo, Janssen, MSD, Novartis, Pfizer, and Sanofi; and personal participation on a data safety monitoring or advisory board for Arvinas, CureVac, Macrogenics, Orion.

Data sharing

Novartis is committed to sharing patient-level data and supporting clinical documents from eligible studies with qualified external researchers. Requests are reviewed and approved by an independent review panel on the basis of scientific merit. All data provided are anonymised to respect the privacy of patients who participated in the trial, in line with applicable laws and regulations. Data availability for this trial is according to the criteria and process described at ClinicalStudyDataRequest.com

For more on **data availability criteria and processes** see https://www.clinicalstudydatarequest.com/

Acknowledgments

This trial was funded by Novartis. MJM's contribution to the study was supported by the NIH/NCI Cancer Center Support Grant P30 CA008748. Under the direction of the authors, Katie Willetts (Oxford PharmaGenesis, Oxford, UK) provided medical writing assistance with funding from Novartis. We thank the participants and their families and caregivers; the staff involved in the trial; and Lamis Eldjerou (formerly of Novartis) for reviewing the manuscript for accuracy.

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